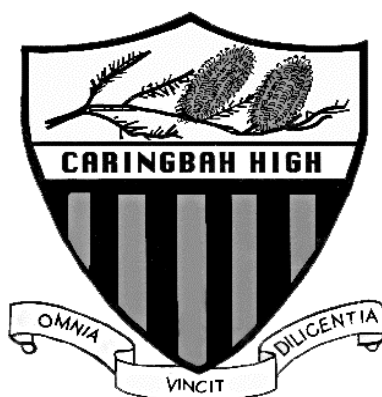




Caringbah High School

Year 11 2019

Mathematics Advanced

Preliminary Course

Assessment Task 4

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-17, show relevant mathematical reasoning and/or calculations
- Answer each question in a separate answer booklet

Total marks – 85

Section I 10 marks

Attempt Questions 1-10

Mark your answers on the answer sheet provided. Detach the sheet and write your name on it.

Section II 75 marks

Attempt Questions 11-17

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only									
Section I	Section II							Total	
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16	Q17		
/10	/11	/11	/11	/11	/11	/10	/10	/85	%

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. Simplify fully $\frac{x^6}{y^6} \div x^2 y^{-3}$?

(A) $\frac{x^3}{y^2}$

(B) $\frac{x^4}{y^2}$

(C) $\frac{x^4}{y^3}$

(D) $\frac{x^4}{y^9}$

2. What is the minimum value of the expression $x^2 + 4x - 1 = 0$?

(A) -1

(B) -5

(C) -9

(D) -13

3. What are the solutions of the equation $\left| \frac{x}{2} - 1 \right| = 3$?

(A) $x = -4$ or $x = -8$

(B) $x = 4$ or $x = -8$

(C) $x = -4$ or $x = 8$

(D) $x = 4$ or $x = 8$

4. The discrete random variable X denotes the number on the upper face when a fair die is rolled once. What is the probability that $4X + 1$ is a perfect square?
- (A) $\frac{1}{6}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{3}$
5. What is the solution of the equation $\tan^2 x - 3 = 0$ for $0 \leq x \leq \pi$.
- (A) $x = \frac{\pi}{6}$ only
- (B) $x = \frac{\pi}{3}$ only
- (C) $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$
- (D) $x = \frac{\pi}{3}$ or $x = \frac{2\pi}{3}$
6. Which of the following statements about the graph of the function $y = 2 \sin 4x$ is correct?
- (A) The amplitude is 2 and the period is $\frac{\pi}{4}$
- (B) The amplitude is 2 and the period is $\frac{\pi}{2}$
- (C) The amplitude is 4 and the period is $\frac{\pi}{2}$
- (D) The amplitude is 4 and the period is π

7. If $a = 10^x$, which of the following is an expression of $\log_{10} \sqrt{a}$?
- (A) $\frac{x}{2}$
- (B) x
- (C) $\sqrt{x}$
- (D) x^2
8. The lines $2x - ky = 0$ and $x - 3y = 0$ are perpendicular. What is the value of k ?
- (A) $-\frac{1}{6}$
- (B) $-\frac{2}{3}$
- (C) $-\frac{3}{2}$
- (D) -6
9. Which of the following is the derivative of $(x^3 + 4)^3$?
- (A) $3(x^3 + 4)^2$
- (B) $3(3x^2 + 4)^2$
- (C) $9x^2(x^3 + 4)^2$
- (D) $9x^2(3x^2 + 4)^2$

10. A function is given by $f(x) = \sqrt{9 - x^2}$. What is the domain of $f(x)$?
- (A) $x \geq 3$
 - (B) $x \leq 3$
 - (C) $-3 \leq x \leq 3$
 - (D) $-9 \leq x \leq 9$

End of Section I- Multiple Choice Questions

Section II

60 marks

Attempt Questions 11–17

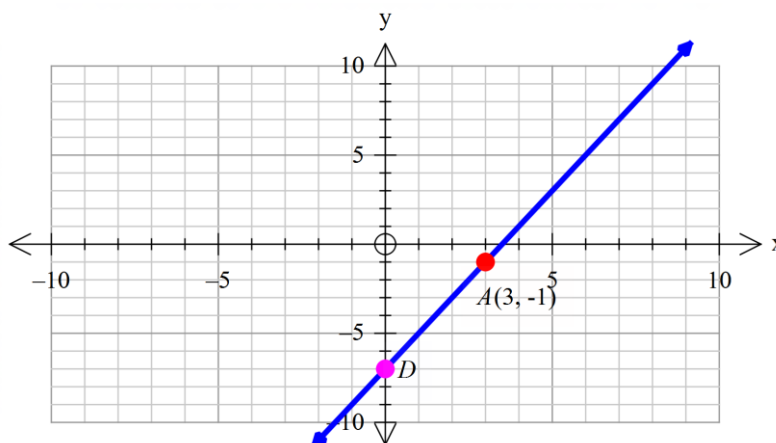
Allow about 1 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (11 marks) Use a SEPARATE writing booklet.

- (a) $A(3, -1)$ is a vertex of the rectangle $ABCD$ whose side AD lies on the line, as shown in the diagram, has an equation $2x - y - 7 = 0$. Find, in general, form the equation of the line that AB lies on. 3



- (b) A fair die is rolled once. A is the event *the score is even* and B is the event *the score is 5 or 6*. Determine, giving reasons, whether the events A and B are independent. 2
- (c) If $f(x) = \sqrt{4-x}$ and $g(x) = x^2$ find
- (i) the composite function $f(g(x))$. 1
 - (ii) the domain and range of $f(g(x))$. 2
- (d) Solve $2\sin^2 x + \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$ 3

End of Question 11

Question 12 (11 marks) Use a SEPARATE writing booklet.

- (a) Solve $\log_x 8 = 1.5$ **1**
- (b) In Year 7 at Random High School, every student must do Art or Music. In a group of 100 students surveyed, 47 do Music and 59 do Art.
- (i) Draw a Venn Diagram to represent the above information. **1**
- (ii) Find the probability that this student does both Music and Art. **1**
- (iii) A student is selected at random. Find the probability that this student does Art only. **1**
- (c) Find any values of x where the graph of $f(x) = \frac{1}{x^2 - 4x}$ is not continuous. **2**
- (d) Sketch the graph of the curve $y = \log_2(x+1)$ showing the intercepts on the axes and the equation of the asymptote. **3**
- (e) In a game of chance, Chloe tosses 2 coins. She wins \$10 for 2 heads, \$5 for 2 tails and nothing for a head and a tail. Find the expected value of this game. **2**

End of Question 12

Question 13 (11 marks) Use a SEPARATE writing booklet.

(a) Differentiate:

(i) $\frac{2}{\sqrt{x}} - 3x^2$ 2

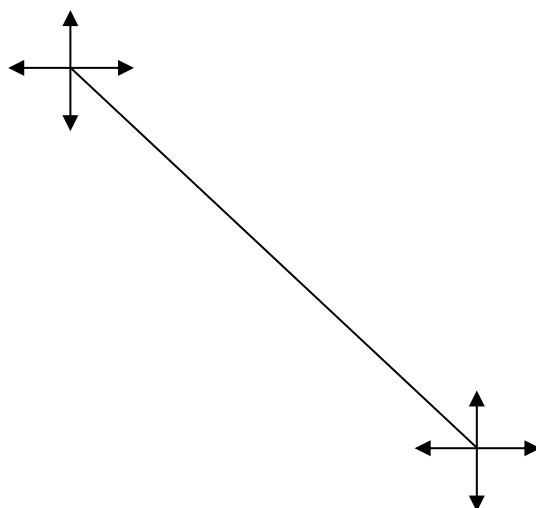
(ii) $(x^2 - 3x)^5$ 2

(iii) $\frac{5 - x^3}{3x - 5}$ 3

(b) Simplify $\frac{16}{2^{3x} \times 8^{1-x}}$ 1

(c) A plane flies from Orange for 1800 km on a bearing of 300° . It then turns and flies for 2500 km on a bearing of 205° .

(i) Represent this question on a diagram by copying completely the diagram below in your booklet. 1



(ii) How far is the plane from Orange, to the nearest km? 2

End of Question 13

Question 14 (11 marks) Use a SEPARATE writing booklet.

- (a) Using first principles formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 + 3$ 2
- (b) Solve the equation $\log_2(x+1) - \log_2(x-1) = 2$ 2
- (c) Using the equation $y = (2x-3)^3$, find
- (i) the equation of the tangent to the curve at the point where $x = 1$. 3
- (ii) the angle of inclination of this tangent line, correct to the nearest degree. 1
- (d) Show that $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = 2 \operatorname{cosec} \theta$ 3

End of Question 14

Question 15 (11 marks) Use a SEPARATE writing booklet.

(a) Sketch the graphs on separate Cartesian planes (one-third page size), showing distinct features and intercepts.

(i) $y = 2^x + 1$ 2

(ii) $y = |x - 2|$ 2

(b) Find the exact value of a and b if $\frac{\sqrt{3} - 4}{2 + 3\sqrt{3}} = a + b\sqrt{3}$. 3

(c) Solve

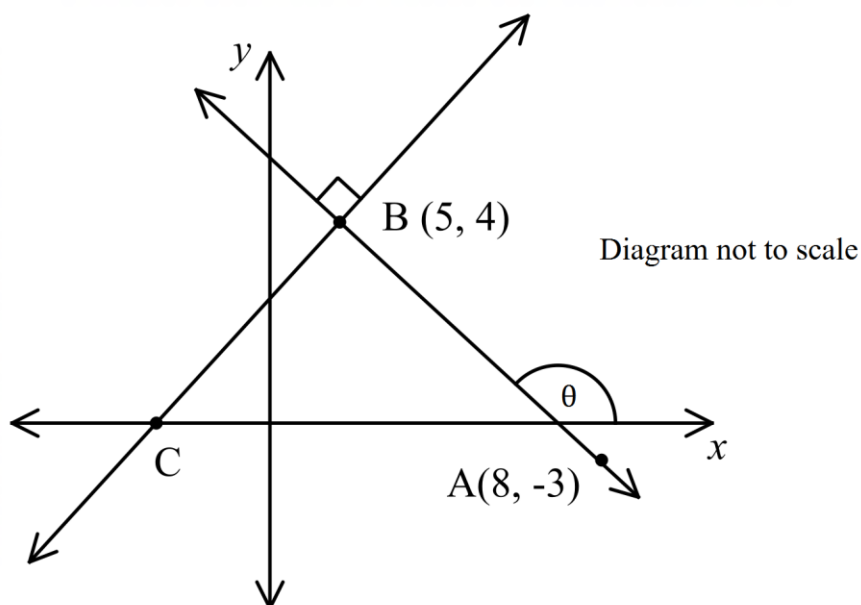
(i) $3x^2 - 5x + 2 \geq 0$ 2

(ii) $|4t - 5| = t + 2$ 2

End of Question 15

Question 16 (10 marks) Use a SEPARATE writing booklet.

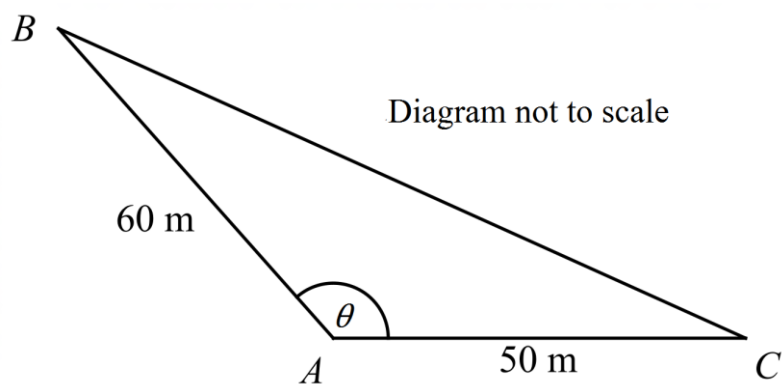
- (a) The points $A(8, -3)$ and $B(5, 4)$ are shown in the diagram below. The line through AB makes an angle of θ with the positive x -axis and the point C lies on the x -axis.



- | | |
|---|---|
| (i) Find the gradient of the line AB | 1 |
| (ii) Find the value of θ to the nearest degree. | 1 |
| (iii) Find the coordinates of C given $AB \perp BC$. | 2 |
| (iv) Find the coordinates of M , the midpoint of AB . | 1 |
| (v) Find the equation of the line, in general form, through M parallel to the line BC . | 2 |

Q16 continued on page 11

- (b) Two sides of a triangular field are $AB = 60$ m, $AC = 50$ m and the included angle $\angle BAC = \theta$ is obtuse.



- (i) If the area of the triangle is 750 m^2 , find the size of $\angle BAC$ **1**
- (ii) Find the length of the side BC , correct to 1 decimal place. **2**

End of Question 16

Question 17 (10 marks) Use a SEPARATE writing booklet.

(a) Consider the circle $x^2 - 2x + y^2 - 4y = 13$

(i) Find the centre and its radius

2

(ii) Show that the line $y = x - 5$ is a tangent to the circle.

3

(b) Consider the function $f(x) = \frac{2x}{x^2 - 1}$.

1

Determine with justification whether the function is odd, even or neither.

(c) A discrete random variable X with mean, $\mu = 2.1$, has probability distribution

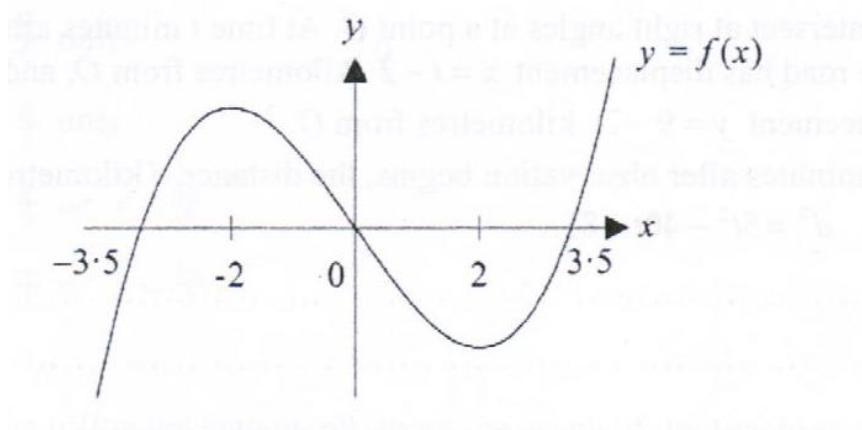
x	1	2	3
$P(X = x)$	0.2	a	b

Find the values of the constants a and b .

3

(d) Using the curve given, sketch $f(-x)$.

1



End of Question 17

End of Examination!

Student Name: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

Yr 11 Advanced Exam 2019 AT4

Multiple Choice

$$\begin{aligned} \textcircled{1} \frac{x^6}{y^6} \div x^2 y^{-3} &= x^6 y^{-6} \div x^2 y^{-3} \\ &= x^{6-2} y^{-6-(-3)} \\ &= x^4 y^{-6+3} \\ &= x^4 y^{-3} \\ &= \frac{x^4}{y^3} \quad (C) \end{aligned}$$

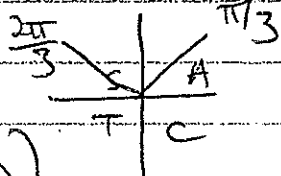
$$\begin{aligned} \textcircled{2} x^2 + 4x - 1 &= 0 \\ x^2 + 4x + 4 &= 1 + 4 \\ (x+2)^2 &= 5 \\ (x+2)^2 - 5 &= 0 \\ \therefore \text{min value is } -5 \quad (B) \end{aligned}$$

$$\begin{aligned} \textcircled{3} \left| \frac{x}{2} - 1 \right| &= 3 \\ \frac{x}{2} - 1 &= 3 \quad \text{OR} \quad \frac{x}{2} - 1 = -3 \\ \frac{x}{2} &= 4 \quad \frac{x}{2} = -2 \\ x &= 8 \quad x = -4 \\ \therefore x &= -4, 8 \quad (C) \end{aligned}$$

④	X	1	2	3	4	5	6
	4x+1	5	9	13	17	21	25
		*					*

$$\begin{aligned} \therefore P(\text{perfect square}) &= \frac{2}{6} \\ &= \frac{1}{3} \quad (B) \end{aligned}$$

$$\begin{aligned} \textcircled{5} \tan^2 x - 3 &= 0 \quad 0 \leq x \leq \pi \\ \tan^2 x &= 3 \\ \tan x &= \pm \sqrt{3} \\ \therefore x &= \frac{\pi}{3}, \frac{2\pi}{3} \quad (B) \end{aligned}$$



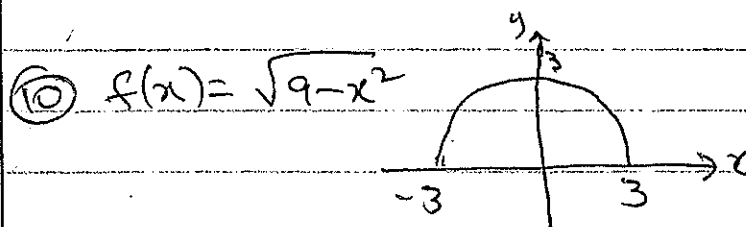
$$\begin{aligned} \textcircled{6} y &= 2 \sin 4x \\ \text{amplitude} &= 2, \text{period} = \frac{2\pi}{4} = \frac{\pi}{2} \quad (B) \end{aligned}$$

$$\begin{aligned} \textcircled{7} a &= 10^x \\ \sqrt{a} &= 10^{x/2} \\ \log_{10} \sqrt{a} &= \log_{10} 10^{x/2} \\ \therefore \frac{x}{2} &= \log_{10} \sqrt{a} \quad (A) \end{aligned}$$

$$\begin{aligned} \textcircled{8} 2x - ky &= 0 \quad x - 3y = 0 \\ ky &= 2x \quad 3y = x \\ y &= \frac{2}{k}x \quad y = \frac{1}{3}x \end{aligned}$$

$$\begin{aligned} \therefore \frac{2}{k}x \times \frac{1}{3} &= -1 \\ \frac{2}{3k} &= -1 \\ 2 &= -3k \\ k &= -\frac{2}{3} \quad (B) \end{aligned}$$

$$\begin{aligned} \textcircled{9} \frac{d}{dx} (x^3 + 4)^3 \\ &= 3(x^3 + 4)^2 (3x^2) \\ &= 9x^2 (x^3 + 4)^2 \quad (C) \end{aligned}$$



$$\text{Domain: } -3 \leq x \leq 3 \quad (C)$$

- ① C ⑤ D ⑨ C mult.
 ② B ⑥ B ⑩ C choice
 ③ C ⑦ A
 ④ B ⑧ B

Question ⑪

(a) AD: $2x - y - 7 = 0$. A (3, -1)

$$y = 2x - 7$$

$$m_{AD} = 2, m_{AB} = -\frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y + 1 = -\frac{1}{2}(x - 3)$$

$$2y + 2 = -x + 3$$

$$x + 2y - 1 = 0$$

(b) $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} = P(A \cap B)$

OR $P(A|B) = \frac{1}{6} \div \frac{1}{3} = \frac{1}{2} = P(A)$

$\therefore A$ & B are independent.

(c) $f(x) = \sqrt{4-x}$ $g(x) = x^2$

(i) $f(g(x)) = f(x^2)$

$$= \sqrt{4-x^2}$$

(ii) Domain: $-2 \leq x \leq 2$

Range: $0 \leq y \leq 2$

(d) $2\sin^2 x + \sin x - 1 = 0, 0^\circ \leq x \leq 360^\circ$

Let $m = \sin x$

$$2m^2 + m - 1 = 0$$

$$(m+1)(2m-1) = 0$$

$$m = -1, \frac{1}{2}$$

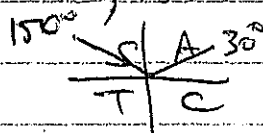
But $m = \sin x$

$$\sin x = -1 \quad \sin x = \frac{1}{2}$$

$$x = 270^\circ$$

$$x = 30^\circ, 150^\circ$$

$$\therefore x = 30^\circ, 150^\circ, 270^\circ$$



Question ⑫

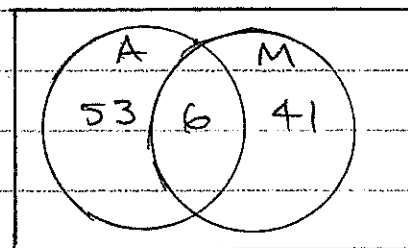
(a) $\log_x 8 = 1.5$

$$(x^{3/2})^{2/3} = (8)^{2/3}$$

$$x = \sqrt[3]{8^2}$$

$$x = 4$$

(b) (i)



$$47 + 59 = 106$$

(ii) $P(\text{both}) = \frac{6}{100} = \frac{3}{50}$

(iii) $P(A \text{ or } M) = \frac{53}{100}$

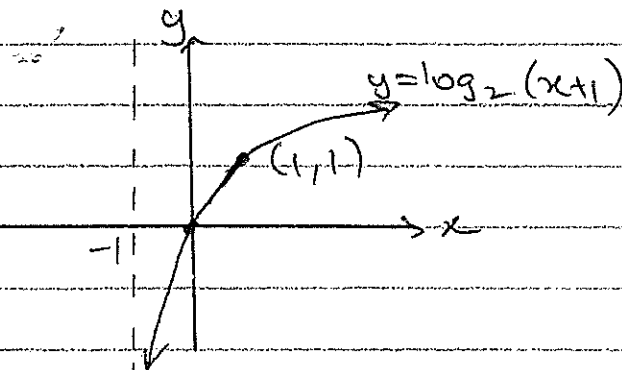
(c) $f(x) = \frac{1}{x^2 - 4x}$

Not continuous: $x^2 - 4x = 0$

$$x(x-4) = 0$$

$$x = 0, 4$$

(d) $y = \log_2(x+1)$



x	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
y	\$0	\$5	\$10
$P(Y=y)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

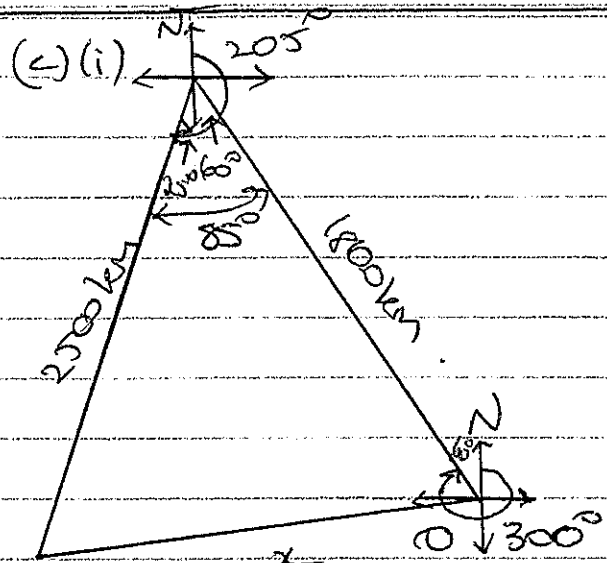
$$E(Y) = \$0 \times \frac{1}{2} + \$5 \times \frac{1}{4} + \$10 \times \frac{1}{4} = \$3.75$$

$$\begin{aligned}
 (13) (a) (i) & \frac{d}{dx} \left(\frac{2}{\sqrt{x}} - 3x^2 \right) \\
 &= \frac{d}{dx} (2x^{-1/2} - 3x^2) \\
 &= -x^{-3/2} - 6x \\
 &= -\frac{1}{\sqrt{x^3}} - 6x
 \end{aligned}$$

$$\begin{aligned}
 (ii) & \frac{d}{dx} (x^2 - 3x)^5 \\
 &= 5(x^2 - 3x)^4 (2x - 3) \\
 &= (10x - 6)(x^2 - 3x)^4
 \end{aligned}$$

$$\begin{aligned}
 (iii) & \frac{d}{dx} \left(\frac{5 - x^3}{3x - 5} \right) \\
 &= \frac{(3x - 5)(-3x^2) - (5 - x^3)(3)}{(3x - 5)^2} \\
 &= \frac{-9x^3 + 15x^2 - 15 + 3x^3}{(3x - 5)^2} \\
 &= \frac{-6x^3 + 15x^2 - 15}{(3x - 5)^2}
 \end{aligned}$$

$$\begin{aligned}
 (b) & \frac{16}{2^{3x} \times 8^{1-x}} = \frac{2^4}{2^{3x} \times 2^{3(1-x)}} \\
 &= \frac{2^4}{2^{3x} \times 2^{3-3x}} \\
 &= \frac{2^4}{2^{3x+3-3x}} \\
 &= \frac{2^4}{2^3} \\
 &= 2
 \end{aligned}$$



$$(ii) x^2 = 2500^2 + 1800^2 - 2(2500)(1800) \cos 85^\circ$$

$$x^2 = 8705598.315$$

$$x = 2950.5$$

$$\approx 2951 \text{ km (nearest km)}$$

Question (14)

$$(a) f(x) = x^2 + 3$$

$$f(x+h) = (x+h)^2 + 3$$

$$= x^2 + 2xh + h^2 + 3$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3 - (x^2 + 3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3 - x^2 - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h}$$

$$= 2x$$

Question 14 cont.

$$(b) \log_2(x+1) - \log_2(x-1) = 2$$

$$\log_2\left(\frac{x+1}{x-1}\right) = 2$$

$$\frac{x+1}{x-1} = 2^2$$

$$\frac{x+1}{x-1} = 4, \quad x \neq 1$$

$$x+1 = 4x-4$$

$$3x = 5$$

$$x = 5/3$$

$$(c) y = (2x-3)^3, \quad x=1$$

$$(i) y' = 3(2x-3)^2(2)$$

$$= 6(2x-3)^2$$

$$\text{when } x=1, \quad y' = 6(2-3)^2$$

$$m = 6$$

$$\text{when } x=1, \quad y = -1$$

$$y - y_1 = m(x - x_1)$$

$$y + 1 = 6(x - 1)$$

$$y + 1 = 6x - 6$$

$$y = 6x - 7$$

$$(ii) m = \tan \theta$$

$$\tan \theta = 6$$

$$\theta = 81^\circ$$

$$(d) \frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = 2 \operatorname{cosec} \theta$$

LHS:

$$\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{(1 + \cos \theta)^2 + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{1 + 2 \cos \theta + 1}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$$

$$= \frac{2}{\sin \theta}$$

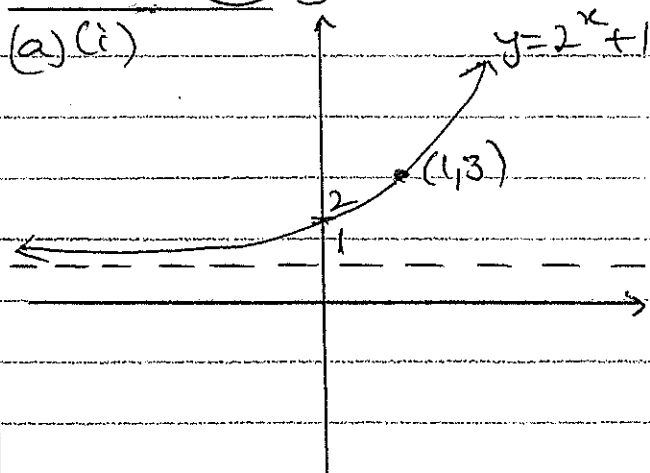
$$= 2 \times \frac{1}{\sin \theta}$$

$$= 2 \operatorname{cosec} \theta$$

$$= \text{RHS}$$

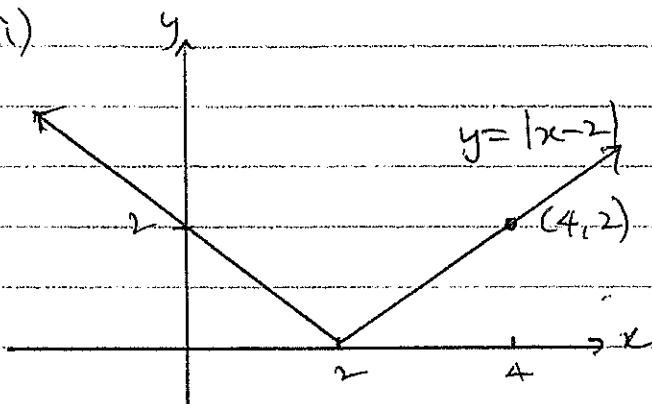
Question 15

(a)(i)



Question (15) cont.

(a) (ii)



$$(b) \frac{\sqrt{3}-4}{2+3\sqrt{3}} = a+b\sqrt{3}$$

LMF:

$$\begin{aligned} & \frac{\sqrt{3}-4}{2+3\sqrt{3}} \times \frac{2-3\sqrt{3}}{2-3\sqrt{3}} \\ &= \frac{2\sqrt{3}-9-8+12\sqrt{3}}{4-27} \\ &= \frac{14\sqrt{3}-17}{-23} \\ &= \frac{-14\sqrt{3}+17}{23} \end{aligned}$$

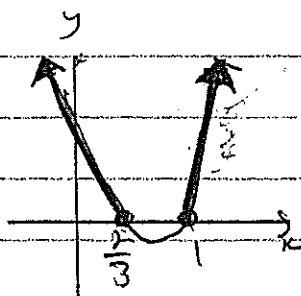
$$\therefore a = \frac{-14}{23}, b = \frac{17}{23}$$

$$(c) (i) 3x^2 - 5x + 2 \geq 0$$

$$(x-1)(3x-2) \geq 0$$

$$x = 1, \frac{2}{3}$$

$$\therefore x \leq \frac{2}{3}, x \geq 1$$



$$(ii) |4t-5| = t+2$$

$$4t-5 = t+2 \quad \text{or} \quad 4t-5 = -(t+2)$$

$$3t = 7$$

$$t = \frac{7}{3}$$

$$4t-5 = -t-2$$

$$5t = 3$$

$$t = \frac{3}{5}$$

$$\therefore t = \frac{3}{5}, \frac{7}{3}$$

Question (16)

$$(a) (i) A(8, -3) \quad B(5, 4)$$

$$m = \frac{4+3}{5-8}$$

$$= \frac{7}{-3}$$

$$= -\frac{7}{3}$$

$$(ii) \tan \theta = -\frac{7}{3}$$

$$\theta = 113^\circ$$

$$(iii) m = \frac{3}{7} \quad B(5, 4)$$

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{3}{7}(x - 5)$$

$$x\text{-int, } y = 0$$

$$-4 = \frac{3}{7}(x - 5)$$

$$-28 = 3x - 15$$

$$3x = -13$$

$$x = -\frac{13}{3}$$

$$\therefore C\left(-\frac{13}{3}, 0\right)$$

$$(iv) A(8, -3) \quad B(5, 4)$$

$$\text{midpoint} = \left(\frac{8+5}{2}, \frac{-3+4}{2}\right)$$

$$= \left(\frac{13}{2}, \frac{1}{2}\right)$$

Question (16) cont.

(a)(v) M $(13/2, 1/2)$ $m = 3/7$

$$y - \frac{1}{2} = \frac{3}{7} \left(x - \frac{13}{2} \right)$$

$$7y - \frac{7}{2} = 3x - \frac{39}{2}$$

$$7y = 3x - 16$$

$$3x - 7y - 16 = 0.$$

(b)(i) $750 = \frac{1}{2} \times 60 \times 50 \sin \theta$

$$750 = 1500 \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

But θ is obtuse, $\theta = 150^\circ$

(ii) $BC^2 = 60^2 + 50^2 - 2(60)(50) \cos 150^\circ$

$$BC^2 = 11296.15242$$

$$BC = 106.3 \text{ m (1 d.p.)}$$

Question (17)

(a) $x^2 - 2x + y^2 - 4y = 13$

(i) $x^2 - 2x + 1 + y^2 - 4y + 4 = 13 + 1 + 4$

$$(x-1)^2 + (y-2)^2 = 18$$

$\therefore$ centre $(1, 2)$, radius $= \sqrt{18}$
 $= 3\sqrt{2}$

(ii) $x^2 - 2x + y^2 - 4y = 13 \dots \textcircled{1}$

$$y = x - 5 \dots \textcircled{2}$$

sub $\textcircled{2}$ into $\textcircled{1}$

$$x^2 - 2x + (x-5)^2 - 4(x-5) = 13$$

$$x^2 - 2x + x^2 - 10x + 25 - 4x + 20 = 13$$

$$2x^2 - 16x + 45 = 13$$

$$2x^2 - 16x + 32 = 0.$$

$$\Delta = b^2 - 4(2)(32)$$

$$\Delta = 0$$

$\therefore$ line is a tangent to the circle.

(b) $f(x) = \frac{2x}{x^2 - 1}$

$$f(-x) = \frac{2(-x)}{(-x)^2 - 1}$$

$$= \frac{-2x}{x^2 - 1}$$

$$= -f(x)$$

$\therefore f(x)$ is an odd fnc.

(c) $0.2 + 2a + 3b = 2.1$

$$2a + 3b = 1.9 \dots \textcircled{1}$$

$$0.2 + a + b = 1$$

$$a + b = 0.8 \dots \textcircled{2}$$

$$\textcircled{1} - 2 \times \textcircled{2}$$

$$2a + 3b = 1.9$$

$$2a + 2b = 1.6$$

$$b = 0.3$$

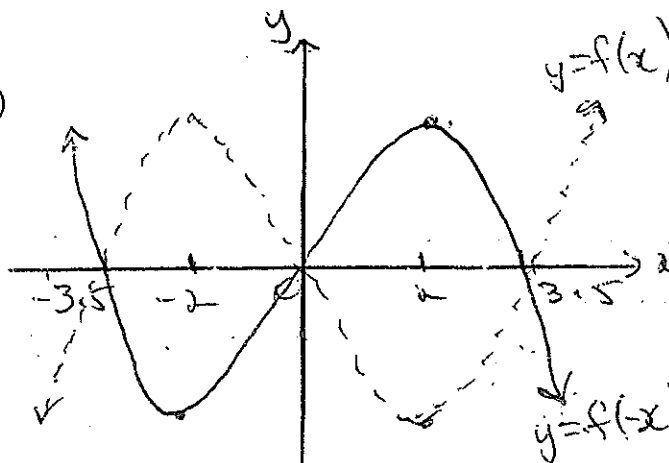
sub $b = 0.3$ into $\textcircled{2}$

$$a + 0.3 = 0.8$$

$$a = 0.5$$

$$\therefore a = 0.5, b = 0.3.$$

(c)



(6)